

Deep Learning-Based Soil Erosion Detection Using Aerial Imagery

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***Abstract* - Soil erosion significantly impacts agricultural productivity, water quality, and ecological balance. This study introduces a deep learning-based framework for detecting soil erosion from aerial (drone or satellite) imagery using a U-Net inspired convolutional neural network. We preprocess a custom dataset of RGB images and binary masks, train a segmentation model using BCE and Dice losses, and evaluate performance through pixel accuracy. Results demonstrate that the model is capable of learning spatial erosion features and producing meaningful segmentation masks. Improvements are discussed through data preprocessing, loss function selection, and training duration.**

1. INTRODUCTION

Soil erosion is a critical environmental issue that leads to the degradation of fertile topsoil, directly affecting agricultural productivity and contributing to water pollution and ecological disturbances. It is primarily driven by natural forces such as wind and water but is often exacerbated by human activities like deforestation, overgrazing, and poor land management practices. Monitoring and managing soil erosion is essential for sustainable land use and food security.

Traditional techniques for detecting erosion typically involve field surveys, manual mapping, and the analysis of satellite imagery through handcrafted features or vegetation indices. However, these methods are labor-intensive and prone to subjectivity. With the advent of machine learning and particularly deep learning, there is a transformative shift toward automated, accurate,

and large-scale analysis of land surface changes. This study leverages a deep learning model based on the U-Net architecture to identify erosion patterns in aerial imagery, enabling efficient and scalable detection.

2. LITERATURE REVIEW

Recent studies have leveraged computer vision and remote sensing for land degradation monitoring. Traditional approaches such as vegetation indices (NDVI), spectral analysis, and GIS-based classification techniques have been useful but are limited by manual intervention and feature engineering. Convolutional Neural Networks (CNNs) have improved land cover classification, yet pixel-wise soil erosion segmentation has been underrepresented. U-Net, originally proposed for biomedical segmentation by Ronneberger et al. (2015), introduced encoder-decoder architecture with skip connections, enabling efficient localization with context. Adaptations of U-Net and other Fully Convolutional Networks (FCNs) have since been applied to agricultural mapping, flood zone detection, and land cover segmentation. Our study draws from these foundations and tailors them for erosion segmentation using high-resolution aerial images.

3. DATASET AND PREPROCESSING

The dataset comprises aerial RGB images and corresponding binary masks indicating erosion. It is structured into train/, test/, and masks/ folders.

Images are resized to 256x256 and normalized. Masks are converted to binary tensors.

4. METHODOLOGY

We implement a simplified U-Net architecture with two encoding blocks, an upsampling block, and skip connections. Binary Cross-Entropy with Logits (BCEWithLogitsLoss) is used alongside Dice Loss to improve mask overlap during training.

5. TRAINING SETUP

- Optimizer: Adam (lr=0.0005)
- Loss: BCEWithLogitsLoss + DiceLoss
- Epochs: 30
- Batch size: 4
- Accuracy metric: Pixel-wise accuracy with a threshold of 0.5

5. RESULTS AND EVALUATION

Initial training with 10 epochs showed low accuracy (~3%), but extended training to 30 epochs improved results steadily. Accuracy and loss curves were plotted to monitor convergence. Visualizations confirmed improved mask prediction.

• Sample Visualizations

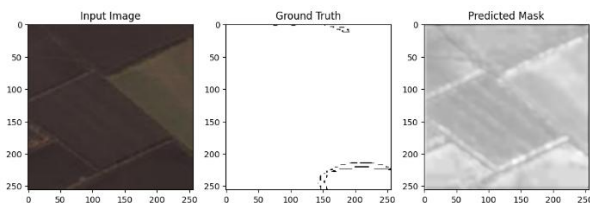


Fig-1

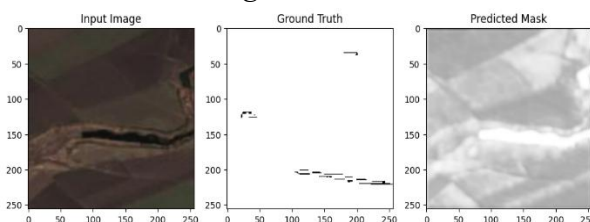


Fig-2

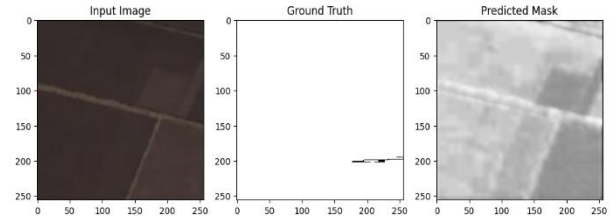


Fig-3

6. DISCUSSION

Challenges included data imbalance and noisy masks. Accuracy was sensitive to loss functions and mask preprocessing. Further improvements can include augmentation, higher resolution inputs, and class balancing.

7. CONCLUSION

This research demonstrates the feasibility of using U-Net based deep learning techniques for automated soil erosion detection from aerial images. Despite initial accuracy limitations, the model showed consistent improvement with extended training. The segmentation results, backed by visual confirmation and accuracy trends, validate the approach. Future work can explore improvements through more advanced architectures (e.g., attention U-Nets), larger datasets, temporal image sequences, and integration with multispectral or hyperspectral data. Real-world deployment would benefit from model generalization across geographies and seasons, enabling authorities and farmers to make timely land management decisions.

8. REFERENCES

- Ronneberger, O. et al. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation.
- Zhang, Y. et al. (2021). Deep learning-based remote sensing for soil erosion detection.
- Long, J. et al. (2015). Fully Convolutional Networks for Semantic Segmentation.

9. APPENDIX

- Code repository:
[<https://github.com/Abhi-1114/projects/blob/52c97870bc0c17e0fdf0c977d0ae39e0c6c32a98/CompVProj2.ipynb>]
- Loss vs Epoch, Accuracy vs Epoch plots

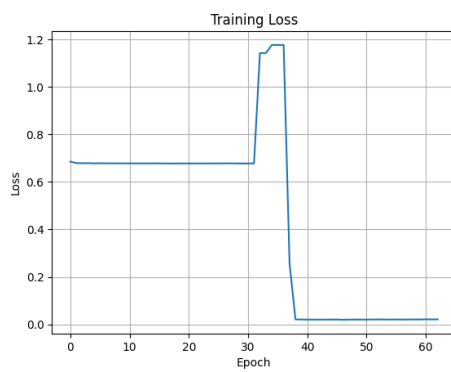


Fig-4

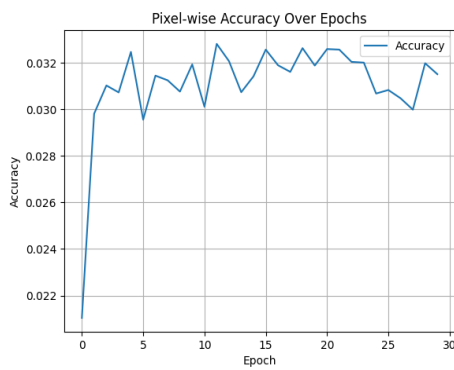


Fig-5